EventPilot: Pre- and Post-STA Analysis Report

Design and Implementation of a Priority-Aware Event Processing Controller Using Verilog HDL

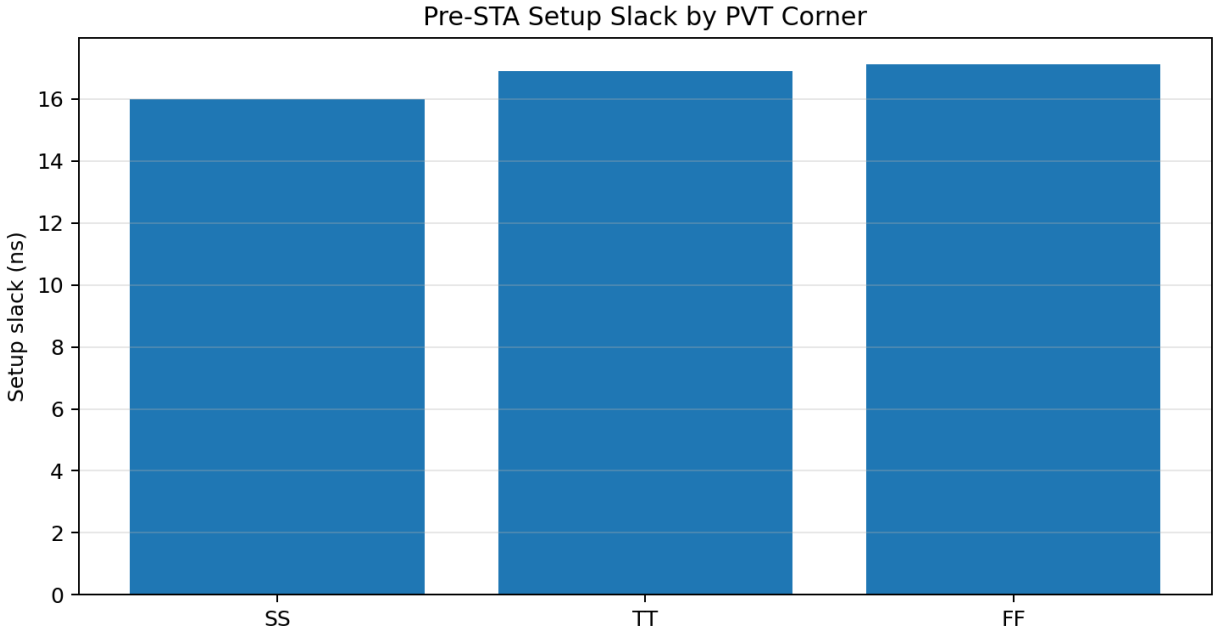
This report visualizes the timing-slack values available in the project context. These are previously recorded values, not newly re-run STA results. Corner and stage labels must be verified against the raw reports before final submission.

1. What STA tells us

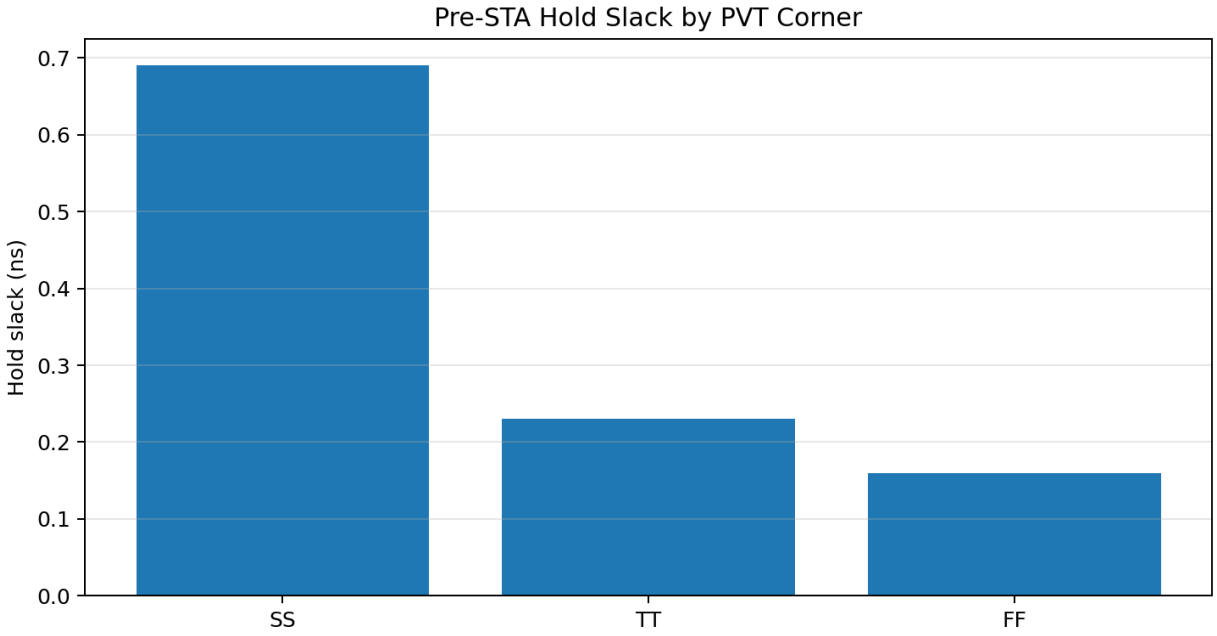
Static Timing Analysis checks whether data paths meet timing constraints. Setup slack measures margin before the capture edge; hold slack measures margin after the capture edge. Positive slack means the reported check passes for the particular constraints, library, path, and parasitic assumptions used. It does not alone prove signoff.

2. Pre-STA: mapped-netlist results

Available pre-STA values are mapped-netlist results without parasitics. SS means slow-slow, TT typical-typical, and FF fast-fast; exact voltage/temperature definitions depend on the PDK setup.



Observation: setup slack is positive in all recorded corners. FF is highest at 17.11 ns and SS lowest at 16.01 ns.



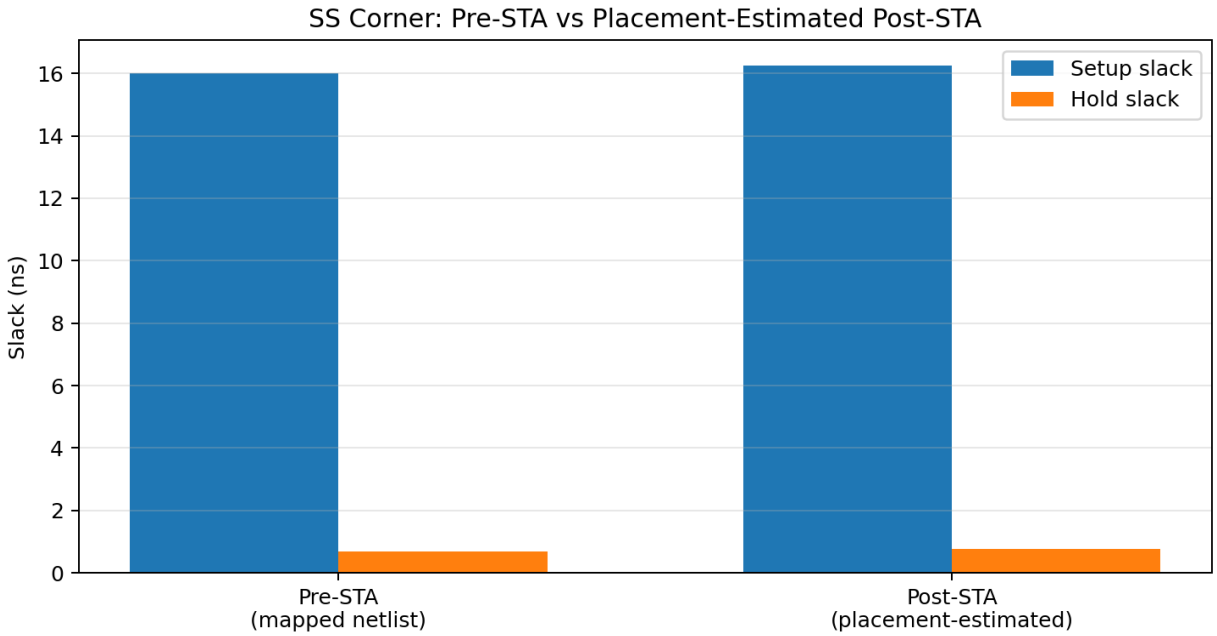
Observation: hold slack is positive in all recorded corners. SS is 0.69 ns, TT is 0.23 ns, and FF is 0.16 ns. FF has the smallest reported hold margin and merits review in the complete path report.

Pre-STA values

Corner	Setup slack (ns)	Hold slack (ns)	Stage
SS	16.01	0.69	Mapped netlist; no parasitics
TT	16.90	0.23	Mapped netlist; no parasitics
FF	17.11	0.16	Mapped netlist; no parasitics

3. SS: Pre-STA vs placement-estimated Post-STA

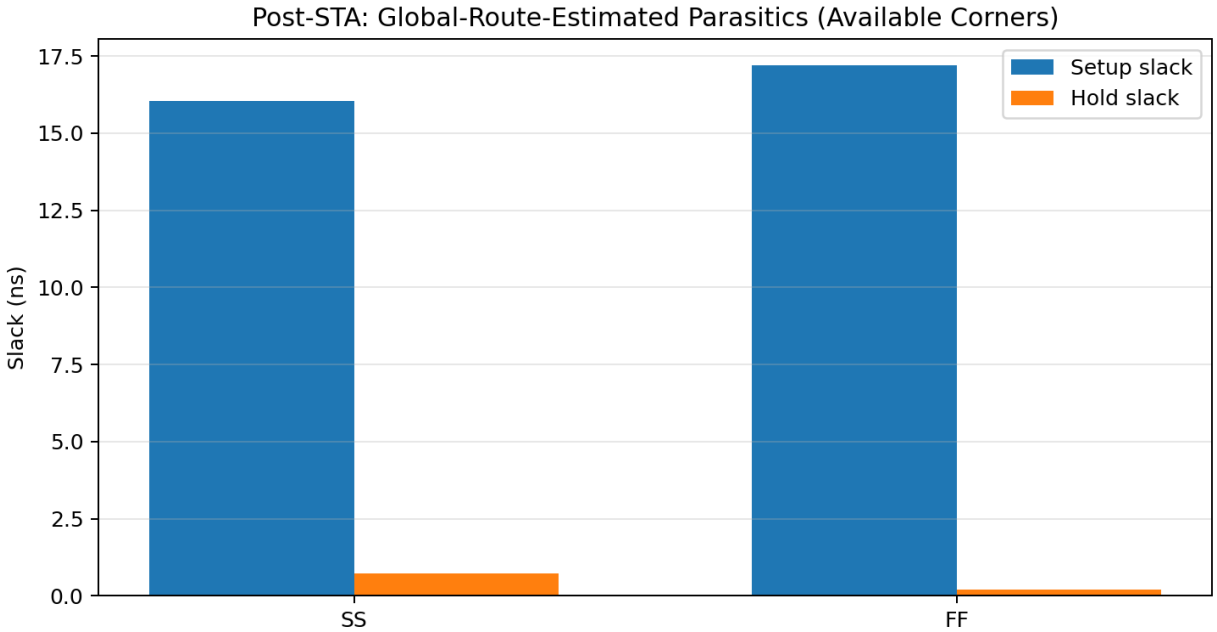
The available placement-estimated SS result is setup slack 16.25 ns and hold slack 0.76 ns. Placement-estimated parasitics are later than a mapped-netlist check but are not the same as final routed/extracted parasitics.



SS comparison: setup slack changes from 16.01 ns to 16.25 ns (+0.24 ns); hold slack changes from 0.69 ns to 0.76 ns (+0.07 ns). Validate these values against the exact report files before final use.

4. Post-STA: global-route-estimated results

Available global-route-estimated results are SS setup 16.05 ns / hold 0.73 ns and FF setup 17.19 ns / hold 0.21 ns. A verified TT global-route result was not available, so it is intentionally not plotted.



Observation: both available corners have positive reported setup and hold slack. SS setup is 0.20 ns lower than the placement-estimated value, and SS hold is 0.03 ns lower. These use different parasitic estimates, so interpret the differences in stage context.

5. Summary table

Stage	Corner	Setup (ns)	Hold (ns)	Note
Pre-STA mapped	SS	16.01	0.69	No parasitics
Pre-STA mapped	TT	16.90	0.23	No parasitics
Pre-STA mapped	FF	17.11	0.16	No parasitics
Placement-estimated	SS	16.25	0.76	Available
Global-route-est.	SS	16.05	0.73	Available
Global-route-est.	FF	17.19	0.21	Available
Global-route-est.	TT	—	—	Not verified/available

6. Interpretation and recommended checks

• Positive slack is encouraging for the listed checks, but does not prove every path or corner passes.
• Verify each raw report’s library corner, timing constraints, path group, and parasitic source.
• Do not label these values final signoff until all required corners and checks are covered.
• Obtain a verified TT post-implementation report if required and run timing with final routed/extracted parasitics when available.
• Include terminal/report screenshots so every plotted number can be traced to evidence.

Data limitation: this PDF was prepared from timing values previously available in the project context and has not independently re-parsed the raw STA logs in this run. Verify the source reports before submitting.